



Green synthesis of gold nanoparticles: Preparation, characterization, cytotoxicity, and anti-bacterial activities

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ARTICLE INFO

Article history:

Received 16 June 2019

Received in revised form 18 July 2019

Accepted 31 August 2019

Available online 31 August 2019

Keywords:

Gold nanoparticles

'Green' synthesis

Olea europaea

Acacia nilotica

Anti-bacterial

Cytotoxic activity

ABSTRACT

Synthesis of nanostructures using green chemistry has been popular owing to the exciting properties and innovative applications of these nanostructures. Novel, cost-effective, and eco-friendly gold nanoparticles (Au NPs) were synthesized using a mix of *Olea europaea* (OE) fruit extract and *Acacia nilotica* (AN) husk extract. The synthesized Au NPs were characterized by UV–visible spectroscopy, and the average NP size (44.96 nm) was measured by a Zetasizer. A more detailed characterization was performed using Fourier transform infrared spectroscopy, scanning, and transmission electron microscopy and electron microscopy with energy dispersive spectroscopy. The cytotoxicity of the synthesized OEAN Au NPs was evaluated using an MTT assay on breast (MCF-7), colon (TCT-116), and hepatocellular (HCEP-2) carcinoma cells. The antibacterial activity of the synthesized OEAN Au NPs was investigated using a disc diffusion method. The synthesized Au NPs exhibited moderate antibacterial activity against the bacterial strains used and profound anticancer activity against different cell lines used.

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1. Introduction

The synthesis of noble metal nanostructures and their applications have been in the forefront of research [1]. Gold (Au) NPs are commonly synthesized using colloidal chemistry methods, which involve using chemicals (reducing/stabilizing reagents) such as sodium citrate, sodium borohydrate, ascorbic acid, and folic acid [2]. Au NPs can be used in the medical field for bio-labeling, diagnosis and detection of various ailments, in biosensor devices for detecting viruses and bacteria, and can act as tiny, precise, and powerful heaters (thermal scalpels) that are able to destroy cancer cells [3]. Recently, green chemistry, which involves the use of several biocompatible methods have been considered for the bio-production of NPs, including methods that use bacteria, fungi, algae, and viruses [3,4]. Some studies have used whole plants,

plant tissues, fruits, plant extracts, and marine algae [5,6]. Thus, green synthesis is environmentally friendly, cost-effective, and most significantly, does not involve high temperatures and pressures, high energy conditions, or the emission of toxic chemical [7].

Hence, the present study, reports a novel and facile method for the biosynthesis of Au NPs with a mixture of *Olea europaea* (OE) fruit extract and *Acacia nilotica* (AN) extract used as a reducing and stabilizing agent. To the best of our knowledge, this is the first study that explores the preparation of Au NPs using a mixture of two different plant extracts (OE and AN). The biosynthetic method used has been granted a patent [8].

2. Material and methods

2.1. Preparation of plant extracts

The OE fruit was obtained from Al-Jouf, Saudi Arabia and crushed and pressed, 10 g of fruit was mixed with 100 mL of distilled water, using a mixer. The extract was then filtered. AN husk that was obtained from Kaboushia, Sudan. To obtain the AN husk

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extract, 1% by volume was soaked overnight in distilled water and then filtered. A mixture of the OE fruit extract and the AN husk extract was prepared, in a 3:5 ratio, respectively, to obtain the final extract solution (OEAN) for the preparation of Au NPs.

2.2. Synthesis of Au NPs

The OEAN aqueous extract was centrifuged for 15 min at 7000 rpm at room temperature, and the resulting pure solution was used for the preparation of NPs. The chloroauric acid was dissolved in 50 mL of distilled water by stirring at 1000 rpm at 40 °C for 15 min. This was followed by the addition of 5 mL of the OEAN extract to the light-yellow aqueous solution of 1 mmol/mL HAuCl₄. A color change to red indicated the reduction of AuCl₄⁻ ions to Au particles, as well as the formation of Au NPs.

2.3. Characterization of green-synthesized NPs

The green-synthesized Au NPs were characterized by UV-visible spectroscopy (UV-visible spectrometer Lambda 25, PerkinElmer, UK). The size of the Au NPs was analyzed using a Zetasizer Nano Series HT laser, (Malvern Instruments, Malvern, UK) and a FT-IR spectrophotometer (Waltham, MA, USA), was used for recording the infrared spectra.

Transmission and scanning electron microscopy (JEOL, Tokyo, Japan) was used for characterizing the size, shape, and morphology of the green-synthesized Au NPs. EDS analysis was performed to confirm the presence of elemental Au.

2.4. Biological assays

The tested human carcinoma cell lines were obtained from the American type culture collection (ATCC, Rockville, MD). A standard MTT cytotoxicity assay was followed. For antimicrobial assay pure cultures of *Escherichia coli*, *Pseudomonas aeruginosa*, *Klebsiella pneumoniae*, *Staphylococcus aureus*, and *Bacillus subtilis* were used to examine the antibacterial activities of the NPs using the disc diffusion method. The analysis was done in triplicates, and the mean zone diameters were determined.

3. Results and discussion

3.1. Characterization

The UV-vis absorption spectrum of the aqueous solution of Au NPs prepared from the OEAN extract, shown in Fig. 1(A), indicates a surface plasmon absorption band at 617.99 nm. This can be attributed to the oscillation of free conduction electrons induced by the absorption of electromagnetic radiation [9] by Au NPs. The synthesized Au NPs had an average size of 44.96 nm, which was also clearly supported by the appearance of a single 100%-intensity peak with a width of 76.47 nm. Similar distribution graph of spherical AuNPs were reported in a previous study in which 90% of nanoparticles showed particle size of 53.8 nm [10] (Fig. 1(B)).

In the surface plasmon resonance analysis, the emergence of a broad peak and a long tail occurred because of the anisotropy of Au NPs, which was clearly confirmed by the TEM images of Au NPs (Fig. 1C). For larger quantities of the extract, the interaction was concentrated, leading to a size reduction of spherical NPs [4,11].

Fig. 2A presents a comparison of the FTIR spectra for the aqueous OEAN extract (Fig. 2A(a)) and the prepared Au NPs (Fig. 2A(b)).

As shown in Fig. 2A(b), the FT-IR bands of green-synthesized Au NPs exhibited peaks at 3276.13 cm⁻¹, 2336.36 cm⁻¹, 2159.87 cm⁻¹, 2031.16 cm⁻¹, and 1635.36 cm⁻¹, which corresponded to the characteristic peaks of OH, O—H, C=C and/or C=C, X=C=Y and/or C—H, and C—C or/and N—H groups, respectively. Based on the comparison between the FT-IR spectra in Fig. 2A(a) and A(b), there was a small shift in the peak position in Fig. 2A(b). A shift in the peak position for AuNPs indicated considerable roles of functional groups such as proteins, phenolics and aromatic compounds in AuNPs formation [4,12]. This clearly indicates the presence of residual plant extract compounds as the reducing and stabilizing (capping) agents for the Au NPs.

Fig. 2B shows the SEM image of the synthesized Au NPs, demonstrating that most of the NPs are spherical, with dispersed irregular forms and some aggregation. While, Fig. 2C shows the EDS spectra of the Au NPs synthesized using the OEAN extract, which confirms the presence of Au in the suspension [4,12].

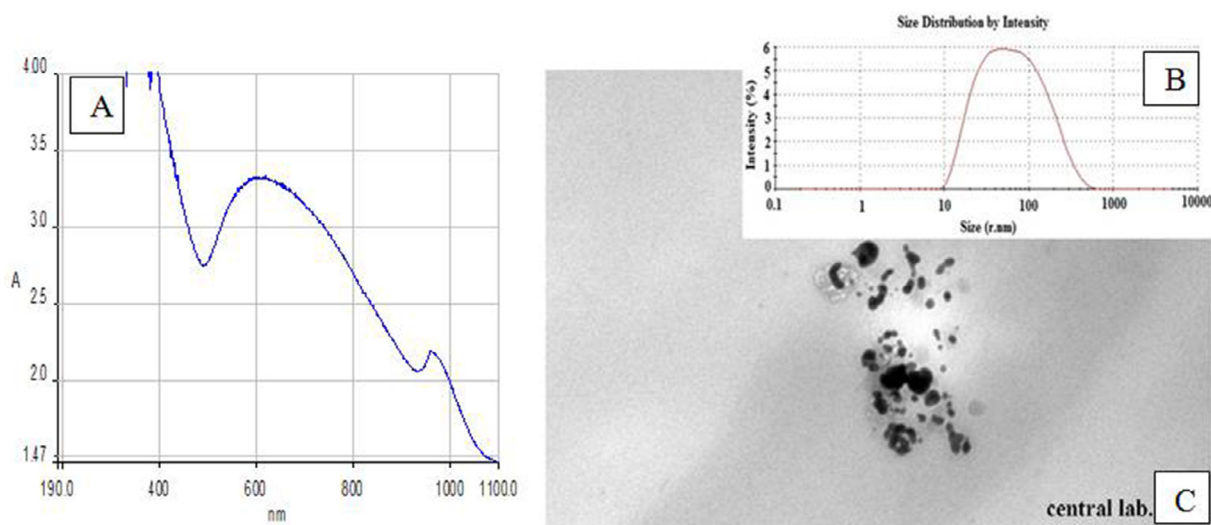


Fig. 1. A) UV-vis absorption spectrum (B) Particle size distribution (C) TEM image for the green-synthesized Au NPs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

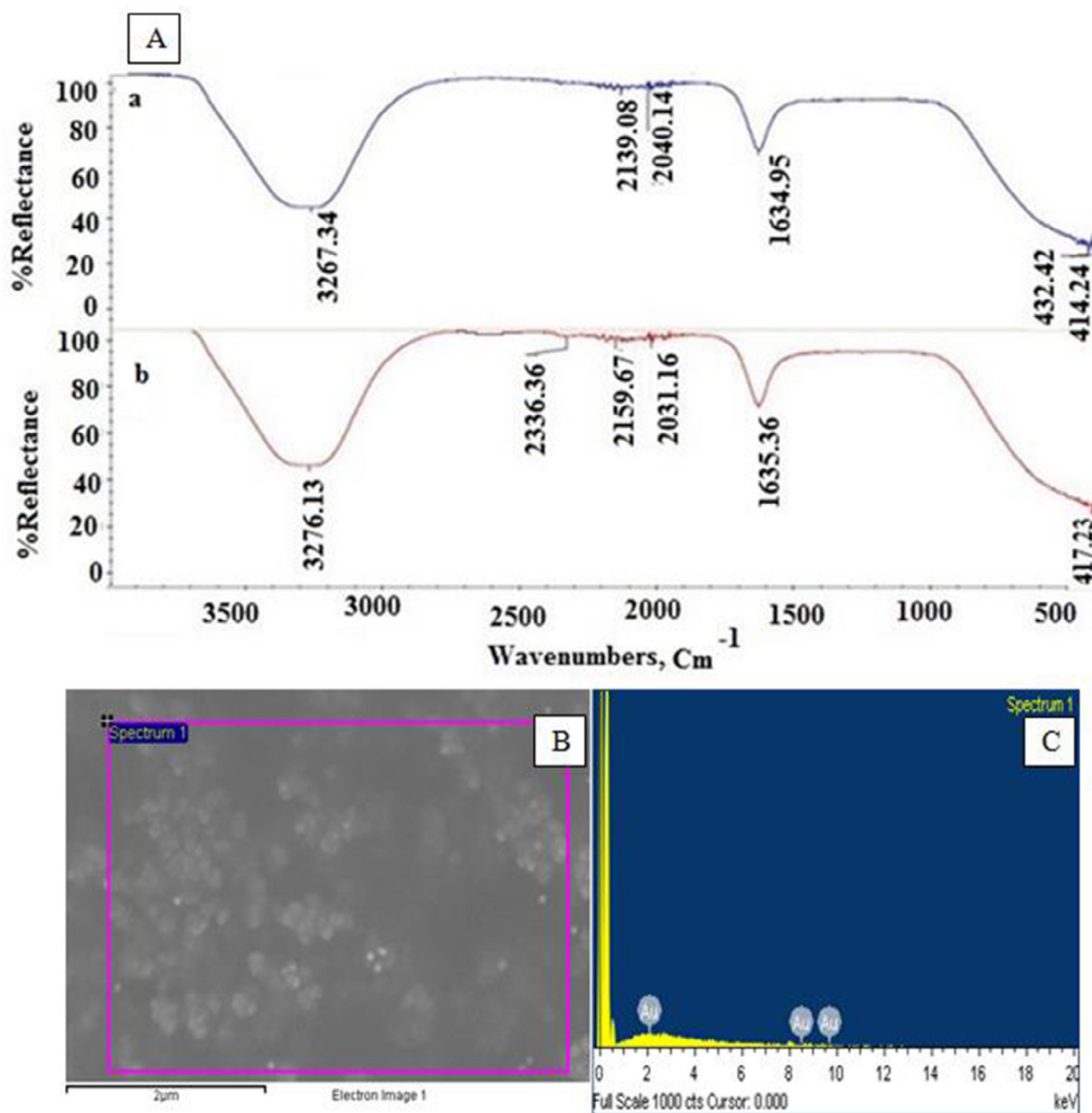


Fig. 2. A) FTIR spectra of (a) the OEAN extract and (b) Au NPs (B) SEM images (C) EDS spectrum for the suspension of green-synthesized Au NPs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.2. Cytotoxicity analysis

The *in vitro* anticancer activity was confirmed by the MTT assay on the cell lines of breast carcinoma cells MCF-7, colon carcinoma cells TCT-116, and hepatocellular carcinoma cells HCepG-2, with IC₅₀ values of the extract above 100 μL for all cell lines and Au NPs. The specific values were IC₅₀ = 45.5 μL for MCF-7 (Fig. 3A), IC₅₀ = 37.2 μL for HCT-116 (Fig. 3B), and IC₅₀ = 40.6 μL for HCepG-2 (Fig. 3C). The Au NPs inhibited the proliferation of cells in a dose- and time-dependent manner [12].

The methods used in the present *in vitro* anticancer study, employed green-synthesized Au NPs on the MCF-7 cell line in a dose-dependent manner, were similar to those used by Prasanna et al. (2009) [13,14]. Similarly, Abel et al. [15] observed the Au NP-capped *C. tora* leaf extract did not affect the viability of normal cells and that the *C. tora* leaf extract conjugated with Au NPs exhibits a higher activity against colon cancer cells.

The cytotoxicity effect of biosynthesized Au NPs against *HeLa* cells was very high (Fig. 3C). Similarly, the cytotoxicity of chemically synthesized Au NPs with respect to *HeLa* cells was reported earlier by [16,17].

3.3. Antibacterial activity study

The suspension of green-synthesized Au NPs and OEAN extract were investigated with respect to their activities against various Gram-positive (*S. aureus* and *B. subtilis*) and Gram-negative (*E. coli*, *Klebsiella pneumoniae*, and *P. aeruginosa*) bacteria. Au NPs demonstrated moderate activity against most of the tested microorganisms, with the highest activity against the Gram-negative bacteria *E. coli*, *K. pneumoniae*, and *Pseudomonas* spp. (Fig. 3D).

El-Tahir et al. [18] performed bioassay-guided fractionation of *A. nilotica* and found that the extract demonstrated the highest activity against *Plasmodium falciparum*.

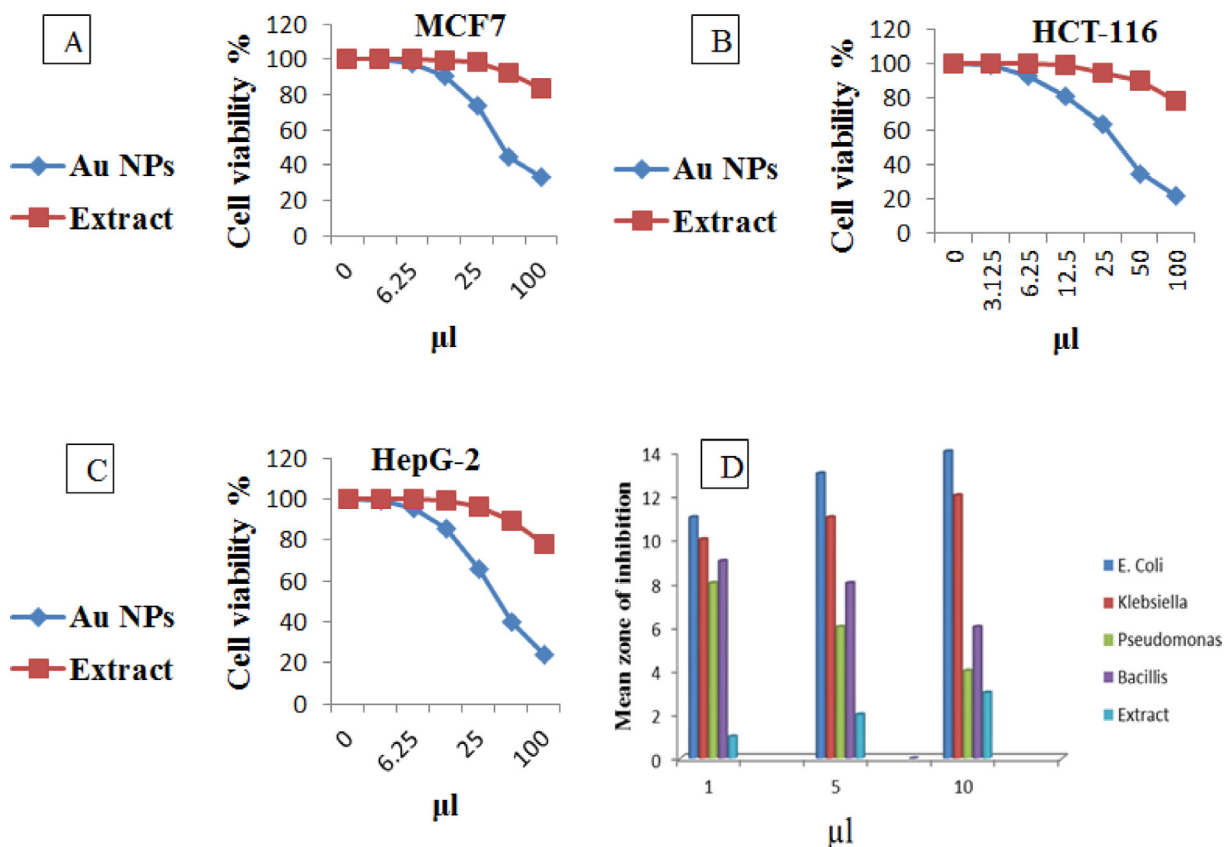


Fig. 3. The Results of the *in vitro* cytotoxicity analysis of synthesized Au NPs of A) MCF-7 cells (B) HCT-116 cells. (C) HepG-2 cells. (D) Antibacterial properties of green-synthesized Au NPs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4. Conclusion

Overall, the results of the study highlighted the green AuNPs promising potential in terms of anticancer and antimicrobial activity. Thus, suggesting a proposed translation of the methodology as a facile, commercially viable and eco-friendly therapeutic approach in the field of nanomedicine/biomedicine.

Author contribution

All co-authors did contribute to this work and are aware of this submission.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research project was supported by a grant from the "Research Center of the Female Scientific and Medical Colleges", Deanship of Scientific Research, King Saud University.

References

- [1] Pareek Vikram, Bhargava Arpit, Gupta Rinki, et al., *Adv. Sci. Eng. Med.* 9 (2017) 527–544.
- [2] R. Aroca, *Surface-Enhanced Vibrational Spectroscopy*, John Wiley & Sons, Chichester, UK, 2006.
- [3] Sahaya Paskalis, Kumar Murphin, Ali Davoodbasha, et al., *Microb. Pathogenesis* 113 (2017) 68–73.
- [4] Chellamuthu Chellapandian, Balakrishnan Ramkumar, et al., *Process Biochem.* 80 (2019) 58–64.
- [5] N. Suganthi, V. Sri Ramkumar, A. Pugazhendhi, et al., *Environ. Sci. Pollut. Res.* 25 (2018) 10418–10433.
- [6] K. Shameli, M. Ahmad, S.D. Jazayeri, P. Shabanzadeh, et al., *Chem. Cent. J.* 6 (2012) 73.
- [7] A. Singh, D. Jain, M.K. Upadhyay, N. Khandelwal, et al., *J. Nanomater. Biosci.* 5 (2010) 483–489.
- [8] Awatif A. Hendi, Manal A. Awad, Nada E. Eisa, Khalid M.O.O. Ortashi, Noble metal nanoparticles, method for preparing the same and their application. Patent. US 9,463,510 B2, Oct. 11, 2016, USPTO.
- [9] M.T. Mirgani, B. Isacchi, M. Sadeghizadeh, F. Marra, et al., *Int. J. Nanomed.* 9 (2014) 403–417.
- [10] Mandana Jafaria, Farrokh Rokhbakhsh-Zamin, et al., *Forootanfa Biocatal. Agric. Biotechnol.* 15 (2018) 245–253.
- [11] F.F. Lucas, H.C. Gustavo, G.S. Jorge, B.L. Ademar, *Nanomaterials (Basel)* 8 (11) (2018) 939.
- [12] Enshirah Da'na, Amel Taha, Eman Afkar, *Appl. Sci.* 8 (2018) 192.
- [13] K.P. Umesh, K.B. Susanta, K.B. Birendra, *Adv. Biol. Chem.* 4 (2014) 360–375.
- [14] R. Prasanna, C.C. Harish, R. Pichai, D. Sakthisekaran, P. Gunasekaran, *Cell. Biol. Int.* 33 (2009) 127–134.
- [15] E.E. Abel, P.R.J. Poonga, S.G. Panicker, *Appl. Nanosci.* 6 (2016) 121–129.
- [16] N. Miura, Y. Shinohara, *Biochem. Biophys. Res. Commun.* 390 (2009) 733–737.
- [17] A. Narsireddy, M. Ranganayaki, B. Anish, M. Meghna, et al., *Int. J. Nanomed.* 10 (2015) 6773–6788.
- [18] A. El-Tahir, G.M.H. Satti, S.A. Khalid, *Phytother. Res.* 13 (1999) 474–478.